# MICHAEL JEFFERSON

## APPLIED AI & FULL-STACK PRODUCT ENGINEER

Vacaville, CA (100% US Remote & SF Bay Area Hybrid) | michael-jefferson-portfolio.vercel.app | github.com/jeffersonlegacy | axion-tax-beyond.vercel.app | tower-hour.vercel.app

Full-Stack TypeScript/JavaScript Engineer and Applied AI Builder specializing in production **Next.js App Router** and **Vite** architectures, **multimodal LLM/Vision and OCR pipelines**, **real-time WebRTC streaming**, and **deterministic agentic workflows**. Proven track record of shipping complex B2B and consumer systems from zero to deployed production—combining deep systems execution with a foundation in cognitive neuroscience and mathematics instruction.

## TECHNICAL STACK

**AI, OCR & Agentic Systems:** Gemini API (Pro / Flash / Vision), Vertex AI, OpenAI SDK & Codex, Intelligent Document Processing (OCR Pipelines, Tesseract.js, ZXing), Ollama (Local LLM Inference), Multi-Agent Orchestration (OpenClaw), Deterministic JSON Schemas, Context Engineering

**Frontend & UI Architecture:** TypeScript, JavaScript (ES6+/ESM), Next.js (App Router, SSR/RSC), React 19, Vite, Tailwind CSS, Chart.js, Remotion (Programmatic Video), H5P Modules, Responsive SPA Design

**Backend, Real-Time & Cloud:** Node.js (ESM), Express, Firebase (Firestore, Realtime Database, Auth, Storage), LiveKit (WebRTC & RBAC JWTs), Twilio 10DLC SMS API & Webhooks, Resend, CloudConvert API, Google OAuth, REST APIs

**DevOps, QA Automation & Systems:** Vercel, Render (render.yaml IaC), Docker, Playwright (Headless E2E & Visual QA), Vitest, Python (ReportLab PDF Pipelines), Sentry Observability, Debian Linux (zram tuning), Tailscale Mesh VPN, 2.5GbE LAN

## SOFTWARE & APPLIED AI ENGINEERING EXPERIENCE

**Lead Full-Stack & Applied AI Engineer — Jefferson Legacy Systems |** Vacaville, CA (Remote / Bay Area) | 2024 – Present

- **Axion Tax & Beyond (axion-tax):** Architected and deployed a decoupled B2B tax intake portal (**Vite + React SPA on Vercel** and **Node.js ESM API on Render via render.yaml**) featuring multi-lane OCR document ingestion, Schedule C & Write-Off Discovery wizards, and bilingual (EN/ES) Twilio SMS + Resend Email alerts.
- **Automated Visual QA & Deliverable Pipeline:** Engineered a headless **Playwright** viewport capture workflow (output/playwright/) paired with a custom **Python PDF synthesis script (build_axion_pitch_deck.py)** to programmatically compile live production UI states into stakeholder pitch decks.
- **Tasty Cuts & More (tasty-cuts-and-more):** Engineered a B2B wholesale meat delivery commerce engine and 48-question Web Audio voice discovery survey in **React 19, Vite 8, and Firebase** with dynamic promo validation (TASTY10 / FILLTHEFREEZER).
- **Tower Hour (tower-hour):** Architected a live-event social commerce and AI product scanner platform in **Next.js App Router, TypeScript, Tailwind CSS, and Firebase** (Firestore + RTDB) across a 373-SKU catalog.
- **WebRTC RBAC & Twilio 10DLC SMS State Machine:** Implemented cryptographically signed **LiveKit RBAC JWTs** isolating broadcaster publish permissions from viewer subscribe tokens, alongside a bidirectional **Twilio 10DLC webhook pipeline** and **Sentry** observability.
- **Multimodal AI Scanner & Local Remotion Worker:** Combined **Tesseract.js, ZXing, and Vertex AI / Gemini** (/api/scan) to map photographed household products to catalog SKU replacements, and built a local **Remotion** render worker for zero-cloud-cost video generation.
- **Jefferson Super Reader 3000:** Built an asynchronous document intelligence pipeline integrating **CloudConvert** and **Gemini Vision APIs** with multi-word chunking, progressive WPM ramp, and a zero-lag **Optimal Recognition Point (ORP)** RSVP engine.
- **Coach Mike AI:** Enforced deterministic **Gemini API JSON response schemas** to drive interactive **Chart.js** psychometric telemetry, Euclidean role alignment, and multi-agent friction coaching matrices.
- **Mike Mac Mesh (mike-mac-mesh):** Provisioned a multi-node local AI mesh linking home devices: headless **Debian Linux** inference server (tuned **zram** swap) with **Ollama** endpoints connected via **2.5GbE LAN** and **Tailscale mesh VPN** to an M2 Mac mini.

**Founder & STEM Systems Instructor — Tower Tutoring |** Vacaville, CA | 2023 – Present

- Architected an interactive collaborative tutoring web platform (**React, Vite, LiveKit WebRTC, Tldraw canvas, OpenAI SDK, Firebase**) and delivered 1-on-1 quantitative coaching, applying cognitive load principles to human-in-the-loop AI UX design.

**Mathematics Instructor & Education Specialist — Public School Districts (EOLA / VUSD / TUSD) |** Solano & Alameda Counties, CA | 2018 – 2025

- Instructed Algebra and Geometry across 7 years and designed data-driven STEM curricula, translating complex quantitative systems into clear mental models for diverse learners and stakeholders.

## EDUCATION

**Sarah Lawrence College** — Bronxville, NY | **Bachelor of Arts** (Psychology & Neuroscience Concentration; Kinetic Sculpture & Physical Computing)

**LaGuardia Community College** — Long Island City, NY | **Associate of Arts**